

FINAL EXPLANATION: CHEMISTRY GRADE 10

Student Assessment Document

FINAL EXPLANATION: CHEMISTRY GRADE 10 — SUB-STRAND 1.2: THE ATOM | FINAL EXPLANATION ASSESSMENT

Student Assessment Document

Student Name	_____
Class	_____
Date	_____

INSTRUCTIONS FOR STUDENTS

WHAT IS THIS DOCUMENT?

This is your Final Explanation Assessment for the unit on The Atom. It is your opportunity to show everything you have learned across the 9 lessons and to fully answer the driving question using evidence, scientific vocabulary, and your own reasoning.

DRIVING QUESTION: Why do different substances produce different colours when burned in a flame, and what does this reveal about how electrons are arranged inside an atom?

HOW TO USE THIS DOCUMENT:

- Read each section prompt carefully before you write.
- Use the exemplar response in each section as a model — notice how it uses evidence from specific lessons, scientific vocabulary, and connects back to the Kenyan cooking hearth phenomenon.
- Your response does NOT need to copy the exemplar. Use your own words, your own evidence, and your own diagrams.
- A strong Final Explanation is evidence-based, well-organised, and shows your thinking — not just memorised facts.
- Refer back to your Driving Question Board (DQB), your lab notebook, and your cumulative atomic model as you write.

WHAT A STRONG RESPONSE LOOKS LIKE:

- ✓ Starts by describing what was puzzling about the phenomenon and what you thought at the beginning
- ✓ Uses specific evidence from lessons (experiments, simulations, readings, discussions)
- ✓ Uses correct scientific vocabulary (atomic number, electron shell, energy level, ground state, excited state, emission)
- ✓ Connects every idea back to the flame colours observed at the Kenyan hearth and in the Kisumu lab
- ✓ Ends with a complete, confident answer to the driving question
- ✓ Includes at least ONE labelled diagram of an atom showing electron energy transitions

LENGTH GUIDE: Aim for 500–800 words across all sections. Quality of reasoning matters more than quantity.

Section 1: What Puzzled Me — Returning to the Phenomenon

Describe the anchoring phenomenon in your own words. What did you observe, and why was it puzzling at the start of the unit? What did you think was happening before you began investigating? Be honest — your starting ideas do not need to have been correct.

At the start of this unit, I thought about something I had actually seen before without understanding it — when my aunt cooks ugali over a wood fire at our homestead near Nakuru, and she accidentally spills salt into the flame, the fire suddenly flashes a bright yellow-orange colour. I always assumed it was just the fire reacting to something wet or oily. I had no idea the colour had anything to do with the atoms inside the salt.

When our teacher introduced the anchoring phenomenon — families in the Rift Valley seeing yellow flames from salt and violet flashes from banana-peel ash — I was genuinely puzzled. My first thought was that the colour came from the physical material burning, like the colour of ash or smoke. I did not understand why sodium always gives yellow and potassium always gives violet, no matter what form they are in. I also did not understand why the colour disappears the moment the substance is removed and comes back when it is added again. That repeatability was the most puzzling part for me. It told me the colour was not random — it was caused by something specific and consistent inside the atom itself. That question — what inside an atom controls the colour of light it produces? — drove my investigation across all nine lessons.

Section 2: The Internal Structure of the Atom — What Atoms Are Actually Made Of

Explain what atoms are made of and describe their internal structure. What are the three main subatomic particles, where are they located, and what charge does each carry? Why does it matter that atoms are mostly empty space? Use evidence from your lessons and include a labelled diagram of an atom.

Before this unit, I thought of atoms as tiny solid balls — the simplest possible building blocks of matter. After our investigations, I now know that atoms have a detailed internal structure that is mostly empty space.

An atom has three types of subatomic particles:

- Protons — positively charged particles found in the nucleus at the centre of the atom.
- Neutrons — particles with no charge, also found in the nucleus alongside the protons.
- Electrons — negatively charged particles that move around the nucleus in specific regions called shells or energy levels.

The nucleus is incredibly small but very dense — it contains almost all of the atom's mass. If you imagined the atom was the size of a large stadium like Kasarani Stadium in Nairobi, the nucleus would be roughly the size of a marble at the centre. All the rest is empty space where the electrons move.

This internal structure is important because it tells us that atoms are not uniform solid objects. They have organised internal parts, and the electrons — being on the outside — are the parts of the atom that interact with energy and with other atoms. This is why electrons are central to explaining flame colours.

[DRAW YOUR LABELLED DIAGRAM HERE: Show the nucleus with protons and neutrons, and electrons arranged in shells around it. Label: nucleus, proton, neutron, electron, shell/energy level.]

Section 3: Atomic Number and Electron Arrangement — What Makes Each Element Unique

Explain what the atomic number is and why it is important. Describe how

Every element is unique because of its atomic number — the number of protons in its nucleus. The atomic number is like an identity card for the element: if you change the number of protons, you get a completely different element. In a neutral atom, the number of electrons is always equal to the number of protons. This means the atomic number also tells us how many electrons the

electrons are arranged in shells around the nucleus, including the rules for how shells fill up. Write the electron configurations for sodium (Na, atomic number 11) and potassium (K, atomic number 19). How does the Periodic Table reflect these patterns? Connect your answer to why sodium and potassium produce different flame colours.	atom has. Electrons do not float randomly around the nucleus. They are arranged in shells — also called energy levels — that fill up in a specific order, from the innermost shell outward: – 1st shell (closest to nucleus): holds a maximum of 2 electrons – 2nd shell: holds a maximum of 8 electrons – 3rd shell: holds a maximum of 8 electrons (for elements 1–20) Using these rules, I can write the electron configurations for the two elements connected to our Kenyan hearth phenomenon: – Sodium (Na, atomic number 11): 2 , 8 , 1 — meaning 2 electrons in the first shell, 8 in the second, and 1 lone electron in the third shell. – Potassium (K, atomic number 19): 2 , 8 , 8 , 1 — meaning a lone electron in the fourth shell. This is already revealing: both sodium and potassium have a single electron in their outermost shell, which is why they are both in Group 1 of the Periodic Table. The Periodic Table is organised by atomic number, and elements in the same group share the same number of outer-shell electrons — which gives them similar (though not identical) chemical behaviour. However, sodium and potassium are NOT identical. Sodium has 11 electrons arranged across 3 shells, while potassium has 19 electrons across 4 shells. That difference in the number of shells and the total electron arrangement means the two atoms interact with energy differently — which is exactly why sodium produces yellow-orange light and potassium produces violet light when heated in a flame.
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Section 4: The Key Explanation — Electron Energy Transitions and Flame Colours

This is the most important section. Explain the complete process by which a flame causes an atom to produce coloured light. Use the terms: ground state, excited state, energy absorption, energy emission, and wavelength. Walk through the process step by step using sodium as your example. Then explain why different elements produce different colours. Include a diagram showing an electron jumping between energy levels.	This section answers the driving question directly. Here is the complete explanation of what happens inside an atom when it is placed in a flame: STEP 1 — HEATING (Energy Absorption): When a substance like table salt (sodium chloride) is placed in the flame of a Bunsen burner or a cooking fire, the atoms absorb thermal energy from the heat. STEP 2 — EXCITATION (Electron Jumps Up): The electrons in the atom use this absorbed energy to jump from their normal, lowest-energy shell to a higher energy shell. We call the normal state the ground state. After absorbing energy, the electron is in an excited state — a higher shell that is further from the nucleus. STEP 3 — INSTABILITY: The excited state is unstable. The electron cannot stay in the higher shell. Within an extremely short time (a fraction of a nanosecond), it falls back down to its original ground state shell. STEP 4 — EMISSION (Light Released): As the electron falls back, it releases the energy it absorbed — but now that energy is released as a packet of light called a photon. The amount of energy released determines the colour (wavelength) of the light: – A large energy drop → high-energy light → blue or violet colour – A medium energy drop → medium-energy light → green or yellow colour
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	– A small energy drop → lower-energy light → orange or red colour For sodium (electron configuration 2,8,1): The lone outer electron in the 3rd shell makes a specific transition that releases energy corresponding to yellow-orange light at a wavelength of approximately 589 nm. This is why every source of sodium — table salt in a Rift Valley cooking fire, sodium street lamps in Nairobi, or a sodium sample in the Kisumu school lab — always glows the same brilliant yellow-orange. For potassium (electron configuration 2,8,8,1): The outer electron is in the 4th shell and makes different transitions with different energy gaps, releasing violet light (~404 nm). This matches the soft violet flash seen when banana-peel ash (rich in potassium) is added to a fire. For copper (electron configuration 2,8,18,1): Copper has more complex electron transitions that produce blue-green light — the colour seen when a copper wire is held in a Bunsen burner flame in the laboratory. The reason each element produces a unique colour is that each element has a unique number of electrons arranged in a unique configuration. Because the shells and the energy gaps between them are different for every element, the energy released during transitions is different, meaning the colour of light emitted is different. Flame colour is therefore a fingerprint of an atom's internal electron structure. [DRAW YOUR DIAGRAM HERE: Show an atom with shells labelled. Draw an electron arrow going UP (energy absorbed from flame) and an arrow going DOWN (photon of coloured light emitted). Label: ground state, excited state, energy absorbed, photon emitted, specific colour of light.]
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Section 5: My Complete Answer to the Driving Question	
Write a full, confident answer to the driving question: 'Why do different substances produce different colours when burned in a flame, and what does this reveal about how electrons are arranged inside an atom?' Your answer should bring together everything from Sections 1–4. Reference the Kenyan cooking hearth phenomenon specifically. Show that you understand both WHAT happens and WHY it happens at the atomic level.	At the start of this unit, I could not explain why the cook at a Rift Valley homestead sees yellow-orange light when salt falls into the fire, or why a neighbour sees violet when burning banana-peel ash. I now have a complete explanation rooted in the internal structure of atoms. Different substances produce different flame colours because every element has a unique arrangement of electrons in shells around the nucleus. When a substance is heated in a flame, its atoms absorb thermal energy. This energy causes the electrons to jump from their ground state — their normal, lowest-energy shell — to an excited state — a higher energy shell. Because electrons are unstable in the excited state, they immediately fall back to the ground state and release the absorbed energy as visible light. The specific colour of that light depends on the size of the energy gap between the shells involved in the transition. Sodium atoms (atomic number 11, electron configuration 2,8,1) produce yellow-orange light because the electron transitions in sodium release energy corresponding to a wavelength of about 589 nm. Potassium atoms (atomic number 19, electron configuration 2,8,8,1) produce violet light because their electron transitions release energy at a different wavelength (~404 nm). Copper produces blue-green light for the same reason — its unique electron arrangement produces unique energy transitions. No two elements have the same electron arrangement, because no two elements have the same number of protons and therefore the same number of electrons. This means every element has its own unique set of possible electron transitions, and therefore its

	own unique flame colour. Flame colour is a visible fingerprint of an atom's electron structure. This reveals something profound about electron arrangement: electrons do not exist at just any energy — they occupy specific, fixed energy levels (shells). They can only absorb or release specific amounts of energy corresponding to the gaps between those levels. This quantised nature of electron energy is what makes each element's flame colour sharp, specific, and repeatable — which is exactly what the cook at the Rift Valley hearth observes every single time salt touches the flame. The phenomenon that seemed like a simple kitchen accident turns out to be a window into the hidden architecture of the atom itself.
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RUBRIC			
Criterion	Excellent (4)	Proficient (3)	Developing (1–2)
1. Explanation of the Phenomenon (Sense-making)	Clearly and accurately describes the anchoring phenomenon (Kenyan cooking hearth / Kisumu lab). Honestly reflects on initial thinking and explains precisely what was puzzling and why prior knowledge was insufficient. Demonstrates a strong sense of intellectual journey from confusion to understanding.	Adequately describes the phenomenon and identifies what was puzzling. Reflects on initial thinking with some clarity. The connection between the phenomenon and the investigation is evident, though not always deeply articulated.	Describes the phenomenon vaguely or only partially. Little or no reflection on prior thinking. The link between the observed colours and the scientific investigation is unclear or missing.
2. Scientific Accuracy — Atomic Structure and Electron Configuration	Accurately describes atomic structure (protons, neutrons, electrons; nucleus; shells). Correctly applies the rules for filling electron shells and writes accurate electron configurations for sodium (2,8,1) and potassium (2,8,8,1) with no errors. Correctly uses atomic number and mass number. Includes a clear, correctly labelled atomic diagram.	Describes atomic structure with mostly correct information and minor errors. Writes electron configurations that are mostly correct (one shell may be slightly off). Atomic diagram is included and mostly labelled correctly.	Contains significant errors in atomic structure or electron configurations. May confuse protons with electrons or apply shell-filling rules incorrectly. Diagram is missing, unlabelled, or inaccurate.
3. Explanation of Electron Energy Transitions (Core Concept)	Accurately and clearly explains the complete process: ground state → energy absorption → excited state → instability → electron falls back → photon emitted. Correctly uses all key terms (ground state, excited state, energy absorption, emission, wavelength/colour relationship). Explains WHY different elements produce different colours (unique electron configurations → unique energy gaps → unique wavelengths).	Explains most of the electron transition process correctly. Uses most key terms accurately. Explains that different elements produce different colours due to different electron arrangements, though the explanation of energy gaps and wavelength may lack depth. Transition diagram may be partially correct.	Explanation of electron transitions is incomplete, vague, or contains significant errors (e.g., states that electrons 'glow' without explaining the transition mechanism). Key terms are missing or misused. Little or no attempt to explain why different elements give different colours at the electron level.

	Includes a correct transition diagram.		
4. Use of Evidence from Lessons	Explicitly references evidence from at least three different lessons or activities (e.g., flame test experiment, electron configuration practice, Periodic Table investigation, atomic model building, simulations). Evidence is used to support claims, not merely listed. Responses are grounded in specific observations and data.	References evidence from at least two lessons or activities. Evidence is mostly used to support claims. Some responses may state evidence without clearly connecting it to the explanation.	Little or no reference to specific lesson evidence. Responses rely on general statements without grounding in observed data or specific activities from the unit.
5. Final Answer — Completeness and Coherence	The final answer (Section 5) is complete, coherent, and directly addresses both parts of the driving question (why colours differ AND what this reveals about electron arrangement). Integrates all key concepts from the unit into a unified explanation. Specifically references the Kenyan cooking hearth phenomenon. Uses precise scientific language consistently throughout.	The final answer addresses the driving question with mostly correct and connected ideas. Both parts of the question are touched upon, though one may be less developed. The Kenyan phenomenon is mentioned. Scientific language is mostly accurate.	The final answer is incomplete, addresses only one part of the driving question, or presents ideas that are disconnected from each other. The Kenyan phenomenon is not referenced or is mentioned only superficially. Scientific language is limited or inaccurate.

ATTRIBUTION AND LICENSING

Curriculum Alignment: Structured in pedagogical alignment with the competence-based curriculum format and learning outcomes established by the Kenya Institute of Curriculum Development (KICD). This material is independently developed and is not officially endorsed or published by KICD.

Content & Pedagogy: Developed and adapted by the educator team at SeaVuria (registered 501(c)(3) in the United States and registered NGO in Kenya, No. 872-850A-BF11), who designed the pedagogical framework, lesson templates and exemplar lessons on which this material is based.

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